



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 2)

2018 HSC Course Assessment Task 1

Wednesday December 6, 2017

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided.

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Ms Lee

☐ 12M4B – Dr Jomaa

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-3	4	5	6	Total	%
MARKS	$\overline{3}$	$\overline{16}$	$\overline{14}$	$\overline{9}$	$\overline{42}$	

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

- ☐ 12M4A – Ms Lee
- ☐ 12M4B – Dr Jomaa
- ☐ 12M4C – Mr Ireland

1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

Section I

3 marks

Attempt Question 1 to 3

Mark your answers on the answer grid provided.

Questions	Marks
1. In which quadrant is the complex number $(-3+3i)^3$ located on the Argand plane? (A) the first quadrant (B) the second quadrant (C) the third quadrant (D) the fourth quadrant	1
2. The equation of the polynomial equation $x^3 - 3x^2 + 2 = 0$ has roots α, β and γ. What is the value of $\alpha^3 + \beta^3 + \gamma^3$? (A) 9 (B) 13 (C) 21 (D) 25	1
3. The polynomial $P(z)$ has real coefficients and $P(0) = -1$. The imaginary number α and the real number β satisfy: $P(\alpha) = 0, \quad P(\beta) = 0, \quad \text{and} \quad P'(\beta) = 0.$ The degree of $P(z)$ is at least: (A) 2 (B) 3 (C) 4 (D) 5	1

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Section II

39 marks

Attempt Questions 4 to 6

Write your answers in the writing booklets supplied. Additional writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 4 (16 Marks)	Commence a NEW page.	Marks
(a) Find $\sqrt{5 - 12i}$.		2
(b) Given that $z = -1 + i$ and $\omega = \sqrt{3} + i$.		
i. Find, in the form $x + iy$, $z\bar{\omega}$.		2
ii. By expressing z and ω in the modulus argument form, show that		3
$\frac{z}{w} = \frac{1}{\sqrt{2}} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right)$		
iii. Hence or otherwise, find the exact values of $\sin \frac{7\pi}{12}$.		2
(c) Sketch the region in the complex plane where the inequalities		3
$ z + 1 - i \leq 2 \quad \text{and} \quad 0 \leq \arg(z + 1 - i) \leq \frac{3\pi}{4} \quad \text{hold.}$		
(d)		
i. Factorise the cubic polynomial $z^3 + 1$ over the field of real numbers.		1
Let ω be one of the non-real roots of the equation $z^3 + 1 = 0$.		
ii. Show that $\omega^2 = \omega - 1$.		2
iii. Hence or otherwise, find the value of $(\omega^2 + \omega + 1)^6$.		1

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Question 5 (14 Marks)

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Marks(a) Let the roots of the polynomial equation $x^3 + 5x - 3 = 0$, be α , β and γ .i. Given that $f(x) = \frac{x^2}{1-x^2}$, find the inverse of $f(x)$, where $0 < x < 1$. **1**ii. Find a cubic polynomial equation which has roots **3**

$$\frac{\alpha^2}{1-\alpha^2}, \frac{\beta^2}{1-\beta^2} \text{ and } \frac{\gamma^2}{1-\gamma^2}.$$

iii. Hence or otherwise, find the value of **1**

$$\left(\frac{\alpha^2}{1-\alpha^2}\right)\left(\frac{\beta^2}{1-\beta^2}\right) + \left(\frac{\alpha^2}{1-\alpha^2}\right)\left(\frac{\gamma^2}{1-\gamma^2}\right) + \left(\frac{\beta^2}{1-\beta^2}\right)\left(\frac{\gamma^2}{1-\gamma^2}\right).$$

(b) If ω is a complex root of $z^7 - 1 = 0$.i. Show that ω is a solution of $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$. **1**ii. Find a quadratic equation with real coefficients having a root $\omega + \omega^2 + \omega^4$. **2**iii. Prove that $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$. **2**(c) Let $P(x) = x^4 + ax^3 + 36x^2 - 35x + b$, where a and b are real constants.Given that $x = 5$ and $x = \frac{1-i\sqrt{3}}{2}$ are zeros of $P(x)$.i. Explain why $x^2 - x + 1$ must be a factor of $P(x)$. **2**ii. Find a and b . **2****Please Turn Over**

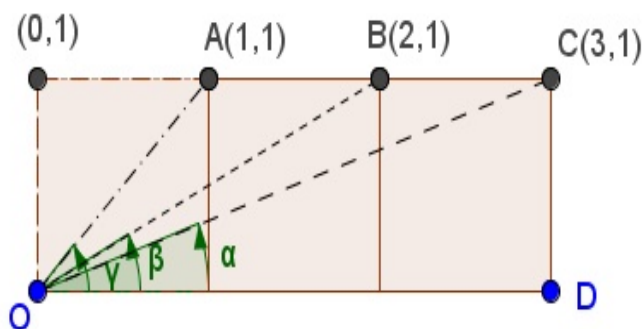
Question 6 (9 Marks)

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Marks

(a)

- i. Prove that $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$.

2

- ii. A , B and C are three points in the Argand diagram (see diagram).

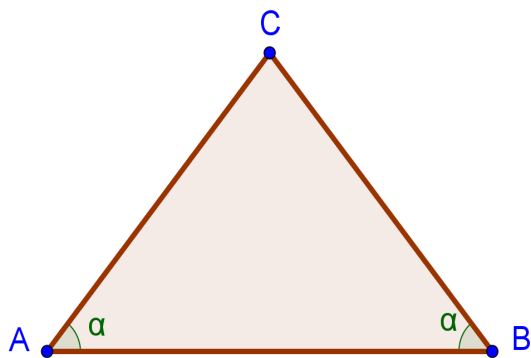
2

Let $\alpha = \angle DOC$, $\beta = \angle DOB$ and $\gamma = \angle DOA$.

By using the diagram above and part (i), show that $\alpha + \beta + \gamma = \frac{\pi}{2}$.

(Do Not find α , β or γ individually).

- (b) In the Argand diagram, the vertices of a triangle ABC are represented by the complex numbers z_1 , z_2 and z_3 respectively. The triangle is isosceles with base AB (see diagram).



- i. Find an expression of $z_3 - z_2$ in terms of $z_2 - z_1$.

2

- ii. Hence or otherwise, show that

3

$$z_3 = \frac{z_2 + z_1 \bar{\omega}}{1 + \bar{\omega}},$$

where $\omega = \cos 2\alpha + i \sin 2\alpha$ and α is the base angle of the triangle.

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$$1. \quad (-3+3i)^3 = 3^3(-1+i)^3 = 3^3 \left(\sqrt{2} \cos \frac{3\pi}{4} \right)^3$$

$$= 54 \sqrt{2} \cos \frac{\pi}{4}$$

in the first quadrant

(A)

$$2. \quad \alpha + \beta + \gamma = 3, \quad \alpha\beta\gamma = -2$$

$$\begin{aligned} \alpha^3 + \beta^3 + \gamma^3 &= 3(\alpha^2 + \beta^2 + \gamma^2) - 2 \times 3 \\ &= 3[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)] - 6 \\ &= 3[9 - 2 \times 0] - 6 \\ &= 27 - 6 \\ &= 21 \end{aligned}$$

(C)

3. $p(z)$ has real coefficients

$P(\alpha) = 0$ and α imaginary

$\therefore P(\bar{\alpha}) = 0$ (Complex conjugate also is a root.)

β is real and $p(\beta) = 0$ and $p(\beta') = 0$

$\therefore \beta$ is a double root

Therefore the degree of $p(z)$ is at least 4.

(C)

4. (16 marks)

$$(a) \sqrt{5-12i} = \sqrt{3^2 - (-2)^2 + 2 \times 3 \times (-2)i}$$
$$= \pm(3-2i)$$

$$(b) z = -1+i, \quad \omega = \sqrt{3}+i$$
$$\bar{\omega} = \sqrt{3}-i$$

$$(i) z\bar{\omega} = (-1+i)(\sqrt{3}-i) = -\sqrt{3}+i+i\sqrt{3}+1$$
$$= 1-\sqrt{3}+(1+\sqrt{3})i$$

$$(ii) z = -1+i = \sqrt{2} \cos \frac{3\pi}{4}$$

$$\bar{\omega} = 2 \cos(-\pi/6), \quad |\omega|^2 = (\sqrt{3})^2 + 1^2 = 4$$

$$\frac{z}{\omega} = \frac{z\bar{\omega}}{|\omega|^2} = \frac{1-\sqrt{3}+(1+\sqrt{3})i}{4}$$

$$\frac{z}{\omega} = \frac{z\bar{\omega}}{|\omega|^2} = \frac{\sqrt{2} \cos(\frac{3\pi}{4}) \times 2 \cos(-\pi/6)}{4}$$

$$= \frac{\sqrt{2}}{2} \cos\left(\frac{3\pi}{4} - \frac{\pi}{6}\right)$$

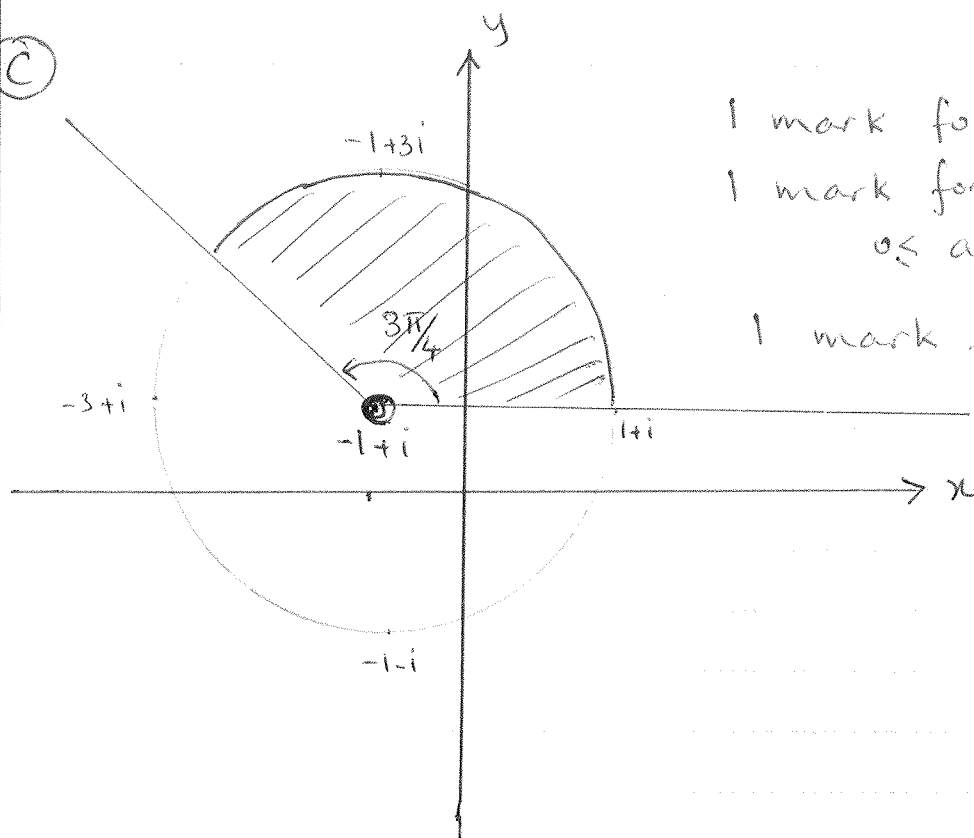
$$= \frac{\sqrt{2}}{2} \cos\left(\frac{7\pi}{12}\right)$$

$$(iii) \frac{1-\sqrt{3}}{4} + \frac{1+\sqrt{3}}{4}i = \frac{\sqrt{2}}{2} \left(\cos\left(\frac{7\pi}{12}\right) + i \sin\left(\frac{7\pi}{12}\right) \right)$$

$$\therefore \frac{\sqrt{2}}{2} \sin \frac{7\pi}{12} = \frac{1+\sqrt{3}}{4}$$

$$\sin \frac{7\pi}{12} = \frac{1+\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{2}+\sqrt{6}}{4}$$

c



1 mark for $|z+1-i| \leq 2$

1 mark for
 $0 \leq \arg(z+1-i) \leq \frac{3\pi}{4}$

1 mark for correct region.

d

(i) $z^3 + 1 = (z+1)(z^2 - z + 1)$ 1

(ii) w is the non real root 1

$\therefore w^2 - w + 1 = 0$ 1

$\therefore w^2 = w - 1$ 1

(iii) $(w^2 + w + 1)^6 = (2w)^6 = 2^6 w^6 = 2^6 = 64$ 1

5. (14 marks)

(a) (i) $f(x) = \frac{x^2}{1-x^2}$

$$\text{let } y = \frac{x^2}{1-x^2} \therefore y(1-x^2) = x^2$$

$$\therefore x^2(1+y) = y$$

$$\therefore x^2 = \frac{y}{1+y}$$

$$\therefore x = \sqrt{\frac{y}{1+y}}$$

$$\therefore f^{-1}(x) = \sqrt{\frac{x}{1+x}}$$

(ii) $\left(\sqrt{\frac{x}{1+x}}\right)^3 + 5\sqrt{\frac{x}{1+x}} - 3 = 0$

$$\left(\frac{x}{1+x}\right)\sqrt{\frac{x}{1+x}} + 5\sqrt{\frac{x}{1+x}} - 3 = 0$$

$$\left(\frac{x}{1+x} + 5\right)\sqrt{\frac{x}{1+x}} = 3$$

$$\left(\frac{x}{1+x} + 5\right)^2 \frac{x}{1+x} = 9$$

$$(6x+5)^2 x = 9(1+x)^3$$

$$36x^3 + 60x^2 + 25x = 9(x^3 + 3x^2 + 3x + 1)$$

$$27x^3 + 33x^2 - 2x - 9 = 0$$

(iii) product of roots two at a time = $-\frac{2}{27}$

(b)

(i) $z^7 - 1 = 0$

$\therefore (z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$

Since $z=1$ is the real solution then the roots of $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ are the non-real roots of $z^7 - 1 = 0$.

(ii) let $\alpha = \omega + \omega^2 + \omega^4$

$\bar{\alpha} = \bar{\omega} + \bar{\omega}^2 + \bar{\omega}^4 = \omega^6 + \omega^5 + \omega^3$

$\therefore \alpha + \bar{\alpha} = -1$

and

$\alpha \bar{\alpha} = (\omega + \omega^2 + \omega^4)(\omega^3 + \omega^5 + \omega^6)$

$= \omega^4 + \omega^6 + \omega^7 + \omega^5 + \omega^7 + \omega^8 + \omega^7 + \omega^9 + \omega^{10}$

$= \omega^4 + \omega^6 + 1 + \omega^5 + 1 + \omega + 1 + \omega^2 + \omega^3$

$= 2$

$\therefore$ The quadratic equation with real coefficients is

$z^2 + z + 2 = 0$.

(iii) let $\omega = \cos \frac{2\pi}{7}$

$\omega + \bar{\omega} = 2 \cos \frac{2\pi}{7}$

$\omega^2 + \bar{\omega}^2 = 2 \cos \frac{4\pi}{7} = -2 \cos \frac{3\pi}{7}$

$\omega^4 + \bar{\omega}^4 = 2 \cos \frac{8\pi}{7} = -2 \cos \frac{\pi}{7}$

$\alpha + \bar{\alpha} = -1 \therefore 2 \left(\cos \frac{2\pi}{7} - \cos \frac{\pi}{7} - \cos \frac{3\pi}{7} \right) = -1$

$\therefore \cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$

(C)

(i) $\alpha = \frac{1-i\sqrt{3}}{2}$ is a root $\therefore \bar{\alpha} = \frac{1+i\sqrt{3}}{2}$ is also a root
(polynomial with real coefficients).

$$\alpha + \bar{\alpha} = 1, \quad \alpha \bar{\alpha} = 1$$

$\therefore$ The quadratic equation with α and $\bar{\alpha}$ as the zeros is $x^2 - x + 1 = 0$

$\therefore x^2 - x + 1$ is a factor of $p(x)$.

(ii) $p(5) = 0 \therefore$

$$625 + 125a + 900 - 175 + b = 0$$

$$\boxed{125a + b + 1350 = 0} \quad (1)$$

$$\alpha = \frac{1-i\sqrt{3}}{2}, \quad \alpha^2 = \left(\frac{1-i\sqrt{3}}{2}\right)^2 = \frac{-1}{2} - i\frac{\sqrt{3}}{2}$$

$$\alpha^3 = -1 \quad \text{and} \quad \alpha^4 = \frac{-1}{2} + i\frac{\sqrt{3}}{2}$$

$$p(\alpha) = 0$$

$$\therefore \frac{-1}{2} + i\frac{\sqrt{3}}{2} - a + 36\left(\frac{-1}{2} - i\frac{\sqrt{3}}{2}\right) - 35\left(\frac{1-i\sqrt{3}}{2}\right) + b = 0$$

$$\therefore \boxed{b - a = 36} \quad (2) \quad \therefore b = 36 + a \quad (3)$$

$$(1) \text{ and } (3) \therefore 126a = -1386 \therefore \boxed{a = -11}$$

$$\text{Use } (3) \therefore b = 36 - 11 = \boxed{25}$$

(6) (9 marks)

(a) (i) let $z_1 = r_1 \cos \alpha$ and $z_2 = r_2 \cos \beta$

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \cos \alpha \cos \beta \\ &= r_1 r_2 (\cos \alpha + i \sin \alpha) (\cos \beta + i \sin \beta) \\ &= r_1 r_2 (\cos \alpha \cos \beta - \sin \alpha \sin \beta + i(\cos \alpha \sin \beta + \sin \alpha \cos \beta)) \\ &= r_1 r_2 (\cos(\alpha + \beta) + i \sin(\alpha + \beta)) \end{aligned}$$

$$\arg(z_1 z_2) = \alpha + \beta = \arg(z_1) + \arg(z_2) \quad |$$

(ii) $z_1 = 1+i$, $z_2 = 2+i$, $z_3 = 3+i$

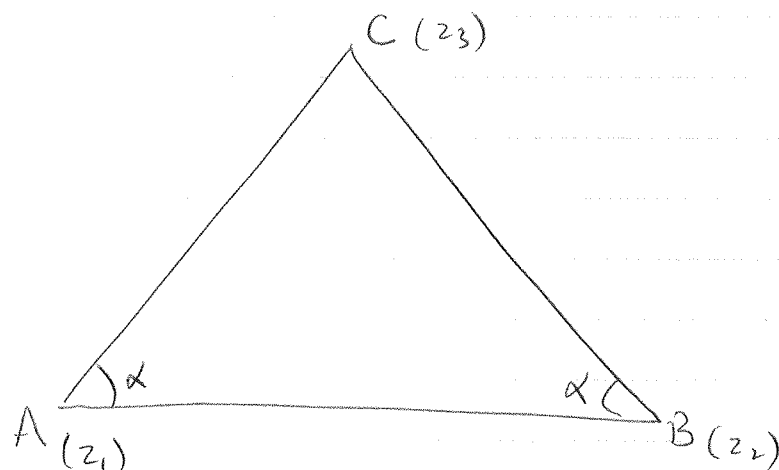
$$\begin{aligned} z_1 z_2 z_3 &= (1+i)(2+i)(3+i) \\ &= (1+i)(6-1+5i) \\ &= 5(1+i)^2 = 10i = 10 \cos\left(\frac{\pi}{2}\right) \end{aligned}$$

$$\arg(z_1 z_2 z_3) = \pi/2$$

$$\therefore \arg(z_1) + \arg(z_2) + \arg(z_3) = \pi/2$$

$$\therefore \alpha + \beta + \gamma = \pi/2$$

(b)



Method 1: (i) $\vec{BC} = k \vec{AB} \cos(180-\alpha)$, k is a positive constant

$$z_3 - z_2 = k (z_2 - z_1) \cos(180-\alpha)$$

(ii) Similarly $\vec{AB} = \frac{1}{k} \vec{CA} \cos(180-\alpha)$

$$z_2 - z_1 = \frac{1}{k} (z_1 - z_3) \cos(180-\alpha)$$

$$\therefore z_3 - z_2 = (z_1 - z_3) \cos(180-\alpha) \cos(180-\alpha)$$

$$z_3 - z_2 = (z_1 - z_3) (\cos 2\alpha - i \sin 2\alpha)$$

$$z_3 - z_2 = (z_1 - z_3) \bar{\omega}$$

$$z_3 (1 + \bar{\omega}) = z_2 + z_1 \bar{\omega}$$

$$z_3 = \frac{z_2 + z_1 \bar{\omega}}{1 + \bar{\omega}}$$

OR

Method 2: (i) $\vec{CA} = \vec{BC} \cos 2\alpha$

$$\therefore \vec{BC} = \vec{CA} \cos(-2\alpha) = \vec{CA} \bar{\omega}$$

$$\vec{CA} = \vec{CB} + \vec{BA}$$

$$\vec{BC} = (\vec{CB} + \vec{BA}) \bar{\omega}$$

$$\vec{BC} (1 + \bar{\omega}) = \vec{BA} \bar{\omega}$$

$$\vec{BC} = \vec{BA} \frac{\bar{\omega}}{1 + \bar{\omega}} \therefore z_3 - z_2 = \frac{\bar{\omega}}{1 + \bar{\omega}} (z_1 - z_2)$$

$$(ii) \quad z_3 = z_2 + \frac{\bar{\omega}}{1 + \bar{\omega}} (z_1 - z_2) = \frac{z_2 + \bar{\omega} z_1}{1 + \bar{\omega}}$$